

However, ML projects have more dependencies as compared to traditional application development. Hence, to reproduce the working code to get the same result again, all these dependencies need to be tracked properly.

- **Lack of standards to package and deploy models:** Another important challenge is the need to have a standard mechanism to package and deploy the models across various environments. The link between the code and the model that produced the actual output needs to be tracked meticulously.
- **Lack of a central store to manage models:** During the development of a machine learning application, various versions of the models are created. The archiving of these versions needs to be handled properly.  
MLflow is designed to address all these pain points and make the machine learning based application development process seamless and efficient. MLflow is an MLOps platform that makes the development and delivery of machine learning projects easy and effective.

## MLflow advantages

The major advantages of MLflow are listed below and illustrated in Figure 2.

- One important advantage of MLflow is its ability to support any machine learning library. It has support for various languages.
- MLflow enables machine learning based applications to run the same way in any cloud.
- MLflow has been designed to support a large number of users. It can support one user as well as thousands of users.
- MLflow is scalable. It can work with Apache Spark to support Big Data.

The power of MLflow can be understood by the way it integrates

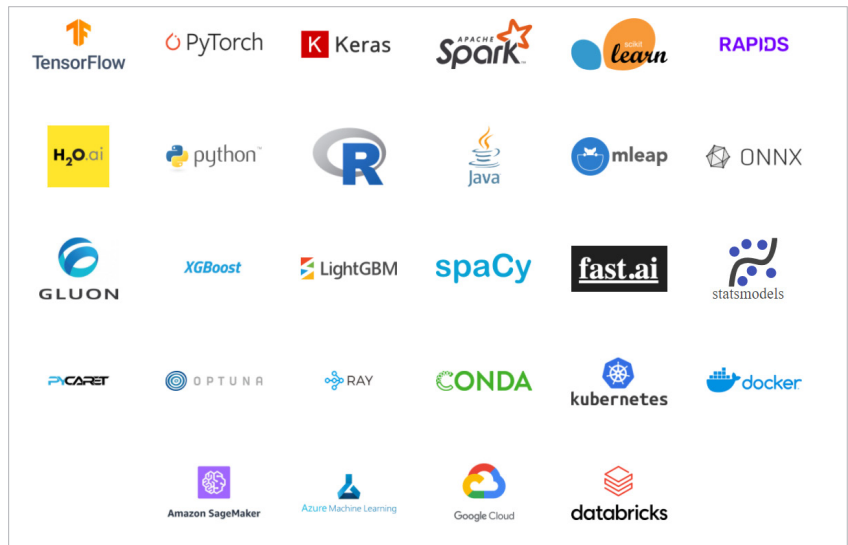


Figure 3: MLflow integrations

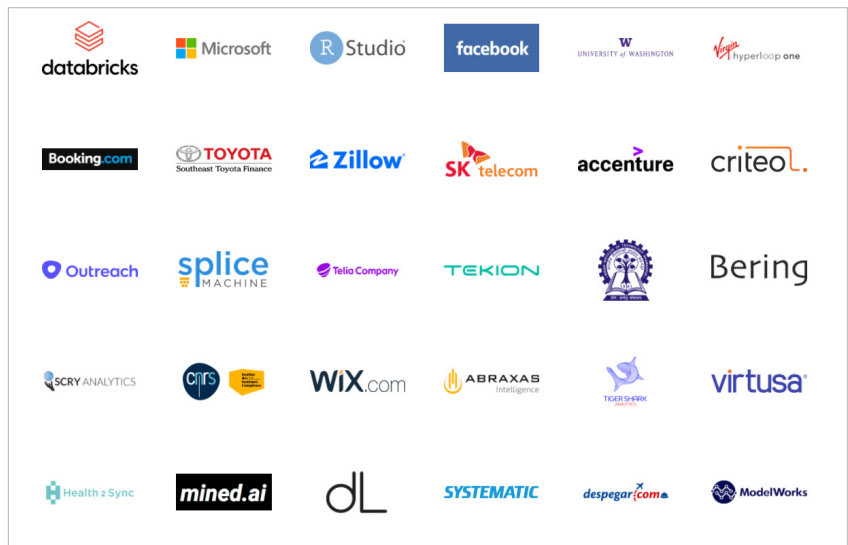


Figure 4: Organisations supporting and using MLflow

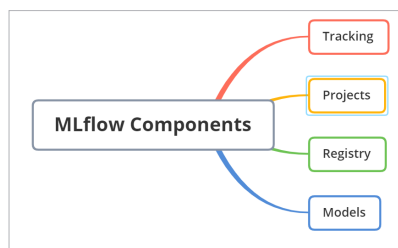


Figure 5: MLflow components

with a large number of ML libraries and languages, as illustrated in Figure 3.

Another proof of the popularity and usability of MLflow is the large

number of big organisations that are using and contributing to it. (A few organisations that support and use MLflow are listed in Figure 4. For the complete list, refer to the official website at <https://mlflow.org/>.)

There are four major components of MLflow (Figure 5):

- MLflow tracking
- MLflow projects
- MLflow models
- Model registry

Now, let's take a quick look at the purpose of each of these components.